Bargaining over Leasing Contracts amid Shifting Power Balance

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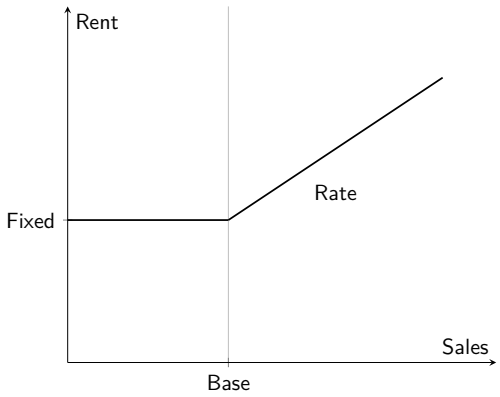
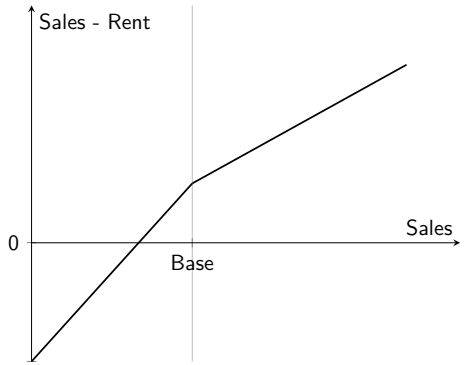
Bargaining over Contract

- Contract specifies how to split the surplus between parties
- Natural (and classical) view: Contract is a target of bargaining
  - Two key elements:
    1. Bargaining set: the set of possible surplus splits
    2. Bargaining power: the power balance between the involved parties
  - Ex: Bargaining over insurance contract (Kihlstrom and Roth, 1982)
- Modern view: Contract theory
  - *take-it-or-leave-it offer* assumption = Extreme power balance
  - Tractable solutions
  - Nice applications: insurance, labor contract, and share cropping

Negotiated Contracts: Leasing Contract in Mall

- Leasing contract specifies *How to split sales between Mall and Tenant*
  - Contract is parameterized by three objects: Fixed, Base, and Rate

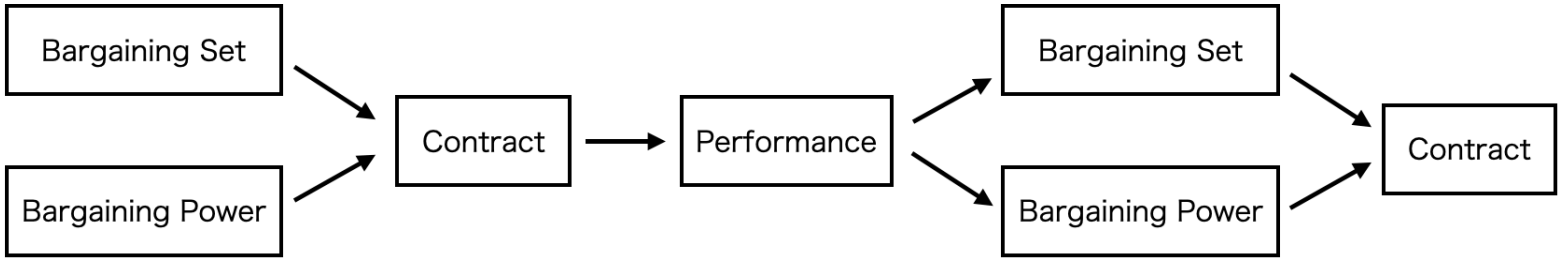




- Intermediate form reflects power balance between Mall and Tenant
  - Best for shopping mall = Fixed rent
  - Best for tenant = Perfect commission
    - High-end store attracts big spenders → Stronger position in negotiation → Advantageous contract (e.g. perfect commission contract)
  - Most contracts take an intermediate form = Mixed-type contract
- More direct observation: Lengthy negotiation (Avg. 3.5 months)
  - Leasing operation remains an individualized skill

Academic Interest: Evolution of Contract

- Evolution of contract = Endogeneous change in contract terms
- Under bargaining view, evolution of contract is decomposed into
  1. Evolution in bargaining set
  2. Evolution in bargaining power
- What drives the evolutions? → Performance during the contract period
  - Ex: Higher sales → Higher expected sales / Stronger bargaining position
- Overview of contract evolution:



This Study

- Focus on leasing contract in shopping mall
- Propose a model and parametrization to capture all the process
  - Bargaining set: Pairs of surplus split between mall and its tenant
  - Bargaining power: Relative strength of mall and its tenant
  - Contract: Monthly rent schedule
  - Performance: Sales from the retail space of tenant
- Empirical findings
  - Bargaining power is not extremely biased
  - Higher sales $\rightarrow$ Stronger positioned tenant
    - About 10% of surplus of mall is driven by the position shift
  - More risk averse $\rightarrow$ Less variant contract
    - Amount of rent can double using less risk-averse contract choice rule

Empirical Challenge: Unobserved Agreement

- Bargaining set is affected by efforts
  - However, efforts are unobserved and also under negotiation
  - Estimation suffers from omitted variable bias
- Two-stage sequential bargaining
  1. Bargaining for effort enforcement
     - Determines the effort and so sales distribution
  2. Bargaining for risk sharing
     - Determines the rent structure given the agreed sales distribution
- Solution concept: (Sequential) Nash-in-Nash solution
- Optimal institutional design supports the separation
  Sales dist. $\rightarrow$ Manager of the mall / Rent structure $\rightarrow$ Representative
  - WHY? $\rightarrow$ In order to avoid overoptimistic contract

Related Literature

- Empirical contract (Lafontaine and Slade, 2013, Lazear, 2018)
  - Relaxing assumptions in empirical analyses (Ackerberg and Botticini, 2002)
  - Empirical contrib.: Risk attitude affects contract terms
  - Empirical contrib.: Bargaining strength affects contract terms
- Empirical bargaining (Lee et al., 2021)
  - Cooperative approach
    - Separating bargaining problem from power makes model tractable
    - Application: Bargaining under externality (Horn and Wolinsky, 1988)
  - Methodological contrib.: Bargaining over complex contract
  - Methodological contrib.: Bargaining over unobserved agreement
- Sources of bargaining powers
  - Theory: Rubinstein (1982, 1985) etc.
  - Recent empirics: Backus et al. (2020)
  - Empirical contrib.: Better performance improves tenant's position

Background

Tenant leasing in a shopping mall

Shopping Mall Management and Renewal Negotiation

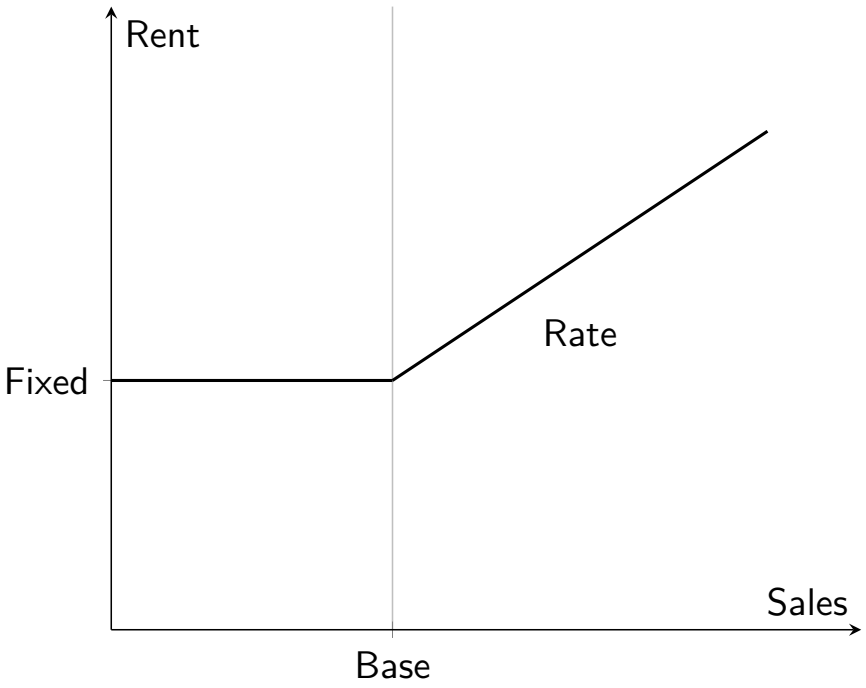
- A shopping mall is operated by a real estate company
  - It generates revenue through tenant rents
- Three phases of shopping mall management:
  1. Searching for new tenants
  2. Negotiating leasing contracts (including renewals)
  3. Maintaining relationships after tenants move in
- Negotiations are delegated to local managers and field staff
  - The company assigns a *manager* to each shopping mall
  - The manager assigns a *representative* to each tenant
- Flow of a renewal negotiation
  1. The mall's initial offer: termination or a renewal proposal
  2. If termination is offered, this decision is non-negotiable
  3. Otherwise, negotiations on the terms of the new lease commence
  4. The tenant begins operations under the new leasing contract

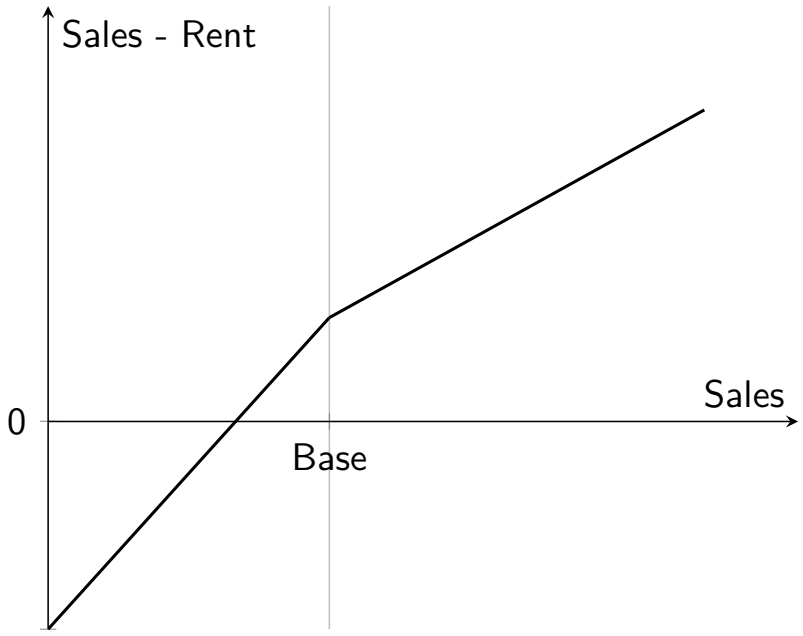
Leasing Contract

- A leasing contract is composed of
  - lease duration, restoration obligations, and monthly rent structure
- Lease durations are typically set to a few years
  - After amortization, the duration is not a big issue for either side
- Restoration obligation is a condition that must be satisfied when exiting
  - A typical example is a skeleton exit
- Rent structure usually becomes a point of conflict. WHY?
  - Professionals say, *"Once expected sales are agreed upon, the rent is naturally determined by industry norms."*
  - The primary obstacle is the uncertainty of future sales (even in renewals)
  - Other conditions are determined by industry standards

Rent Structure

- Monthly rent is typically determined by a variant of share contract
  - Defined by parameters: *Fixed*, *Base*, *Rate*
  - Base = threshold where commission rate changes
  - Rate = commission rate after Base
- Fixed is hard to negotiate → Base plays a role in searching for a compromise





Data

Panel data linking performance and contract terms

Data Source

- Data covers two shopping malls over 6 years (2017-2023)
  - Managed by same company, by different managers
  - Located in same region (Western region of Japan)
  - One in downtown (*Mall 0*), one in suburb (*Mall 1*)
- For them, we have the following two data sources:
  1. performance data
  2. contract data
- They are used for actual leasing operation of the management company

Descriptive Stats

- 226 tenants
- They operate under 443 contracts: Avg. # of renewals is 1.00
- Avg. length of lease duration is 1830 days
- Contract terms vary a lot: All monetary values are scaled by $1,000$ JPY

	Mean	Std	Min	25%	50%	75%	Max
Fixed per Area	5.26	4.11	0.15	2.74	4.51	6.05	27.69
Base per Area	62.02	43.25	3.22	42.36	54.78	60.50	435.57
Rate (%)	0.91	0.30	0.20	0.80	1.00	1.00	3.50

  - Not entirely a matter of the mall's fixed effect Go to common brands comparison

- Commission component is non-negligible

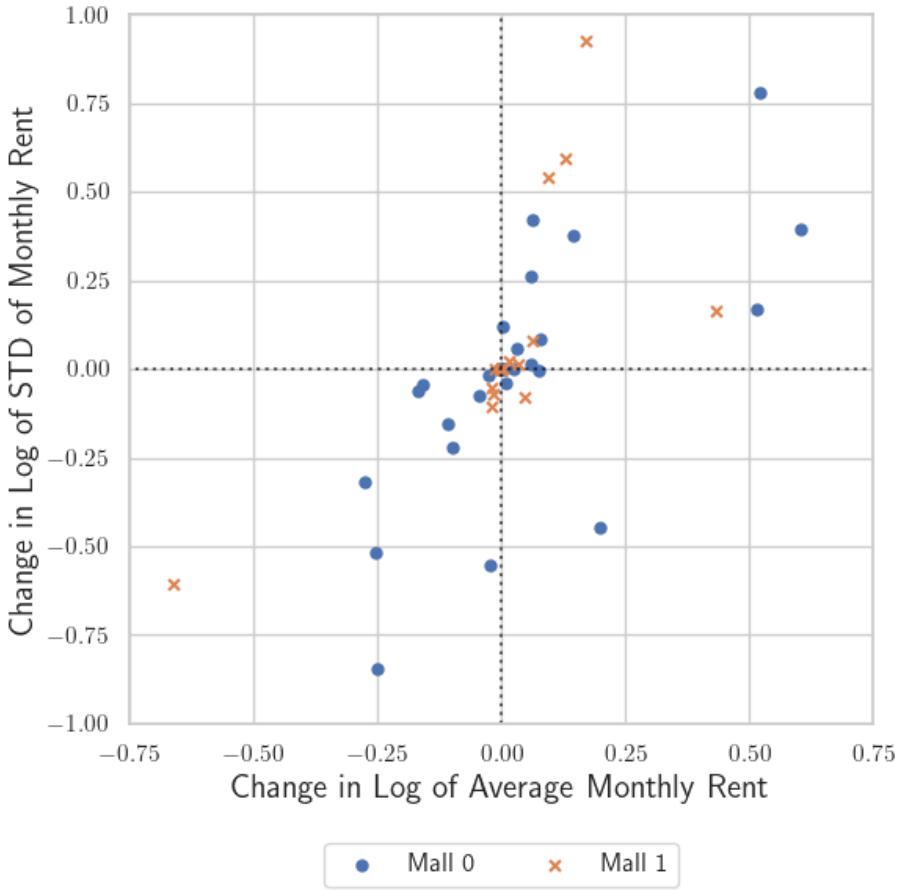
	Mean	Std	Min	25%	50%	75%	Max
Avg. Variable Rent / Total Rent	0.19	0.19	0.00	0.02	0.14	0.31	0.89
Prob. Sales exceeds Base	0.55	0.36	0.00	0.17	0.67	0.88	1.00

Motivational Observations

1. Risk averse agents
2. Past sales affects the outcome of renewal negotiation
3. No agency issue
   - Just fortune simplifying the model
   - If agency issue exists, you can include it in the same frame work
     - See for example Pitchford (1998), Balkenborg (2001), Yao (2012)

Obs. 1 : Risk Averse Agents

- Calculate "Avg. monthly rent" and "Std. of monthly rents" for all contracts
- Compute changes in these values over two successive contracts
- Higher rent must be accompanied by larger variance



Obs. 2: Performance Affects External and Internal Margins

- External margin: Better performance → less exit
- Internal margin: Better performance → more rent

	(1)	(2)	(3)	(4)	(5)	(6)
Dep. Var.	Exit	Exit	Exit	Rent Increase	Rent Increase	Rent Increase
Avg. Sales	-1.50e-09			0.000944*		
	(9.78e-10)			(0.000536)		
Avg. Sales per Area		-1.02e-08			0.00341***	
		(7.19e-09)			(0.000601)	
Avg. Pct. from Bottom			-0.00406***			1258.5***
			(0.000614)			(202.6)
fixed	-5.77e-08**	-5.90e-08**	-3.79e-08	0.161***	0.160***	0.160***
	(2.63e-08)	(2.53e-08)	(2.40e-08)	(0.0253)	(0.0255)	(0.0251)
rate	0.000108**	0.000119**	0.0000360	80.85***	67.87***	94.43***
	(0.0000541)	(0.0000513)	(0.0000499)	(12.01)	(9.701)	(11.99)
area	0.000401	-0.0000225	-0.000190	1560.5***	1802.5***	1918.0***
	(0.000311)	(0.000148)	(0.000169)	(237.2)	(306.5)	(319.8)
N	285	285	285	11820	11820	11820
adj. R^2	0.021	0.018	0.145	0.384	0.383	0.392

Obs. 3 : Little Concern of Moral Hazard

- Regression: Sales $\sim$ Rate + Base + Fixed + Covariates
- A larger fixed rent amount is assigned to tenants with higher sales
  - *Selection* exists
- Sales remain unaffected by contract terms $\rightarrow$ Little concern of moral hazard

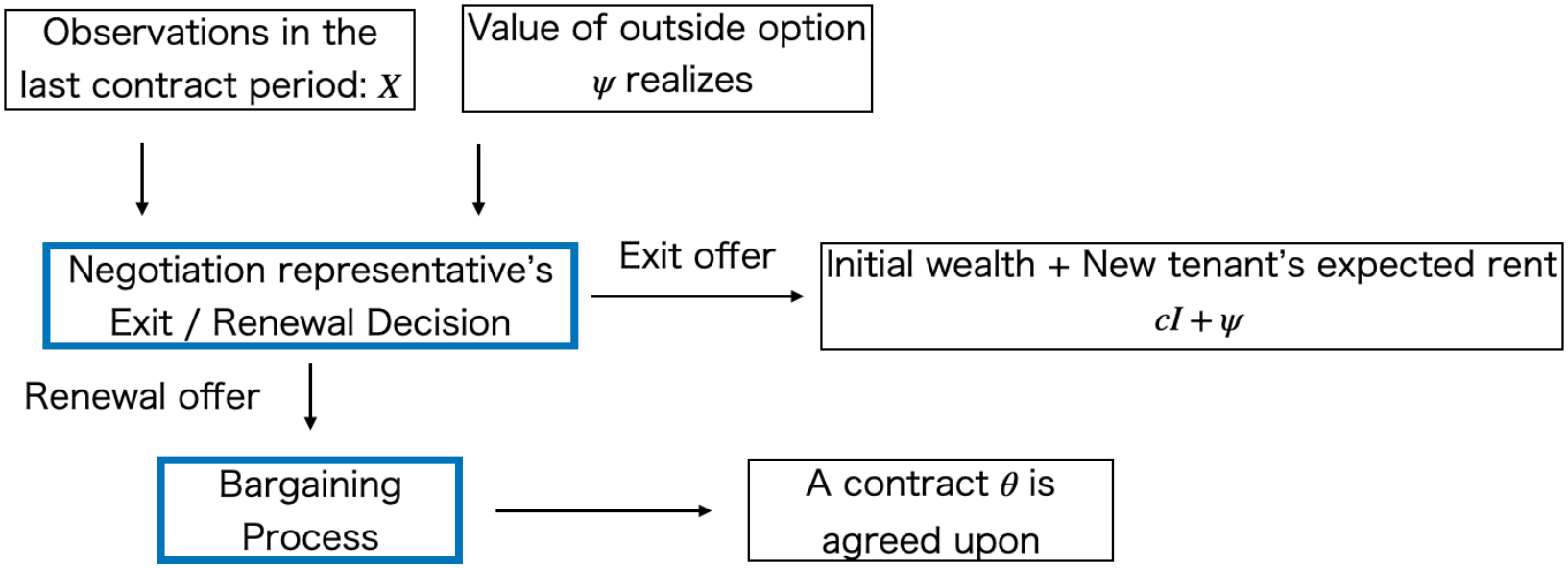
	(1) Level	(2) Diff	(3) Diff / Change
Rate	-0.063	-0.049	-0.180
	(0.219)	(0.222)	(0.374)
Base	0.069	0.000	0.000
	(0.206)	(0.000)	(0.000)
Fixed	2.839**	0.000	0.000
	(1.319)	(0.000)	(0.000)
Observations	197	197	156
Adjusted R^2	0.992	0.964	0.967

Model

Sequential bargaining:

effort enforcement and risk sharing

Model Overview



Go to offer table

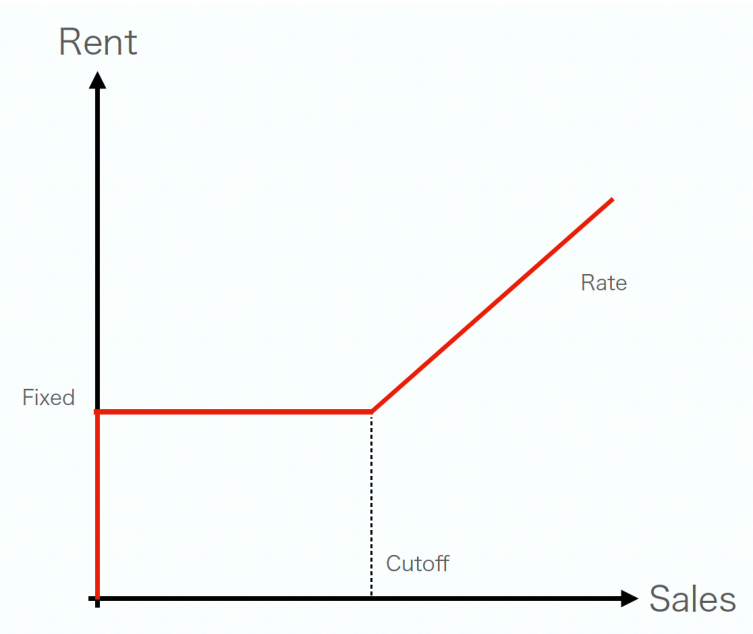
Two-Stage Sequential Bargaining

- Assumption: Sales $S \sim N(\mu, \sigma^2)$ where σ^2 is known
- T: Tenant, M: Manager, R: Representative
- First stage: Bargaining for effort enforcement between M and T
  - Negotiate over μ = Determine managerial effort of M
- Second stage: Bargaining for risk sharing between R and T
  - Negotiation over contract terms given μ
- Solution concept: *Nash-in-Nash*
  - Nash bargaining solution for each bargaining problem
  - First stage is solved in expectation of the second stage result
  - Balance of Bargaining powers: $B = \frac{\text{Bargaining power of Mall}}{\text{Bargaining power of Tenant}}$
    - B^{EE} and B^{RS} for the two bargainings

Rent Structure

- Θ is the parameter space of mixed-type contract: $\theta = (f, b, r) \in \mathbb{R}^3_+$
  - f is Fixed, b is Base, and r is Rate
- The monthly rent is computed as follows

$$R(S; \theta) = f + r \times \max\{0, S - b\}$$



Go to all rent structure tree

Preferences

- Tenant, denote by T

$$U_T(S;\theta) = -e^{-\rho_T(S-R(S;\theta))}$$

- Representative, denoted by R

$$U_R(S;\theta) = -e^{-\rho_R(I+R(S;\theta))}$$

  I is the initial wealth of the shopping mall

- Preferences of R and M are partially mis-aligned due to managerial cost

$$U_M(S,;\theta) = U_R(S;\theta) - C(\mu,\theta,X)$$

- $C(\mu,\theta,X) = \lambda\left(\mathbb{E}_{S\sim N(\mu,\sigma^2)}\left[U_R(S;\theta)\right] - U_R\left(\hat{\mu}(X);\hat{\theta}\right)\right)^2$
  - $\hat{\mu}(X)$ is the observed average sales and $\hat{\theta}$ is the ongoing contract
  - This specification allows for explicit solution $\rightarrow$ Benchmark model

Risk Sharing: Bargaining Frontier

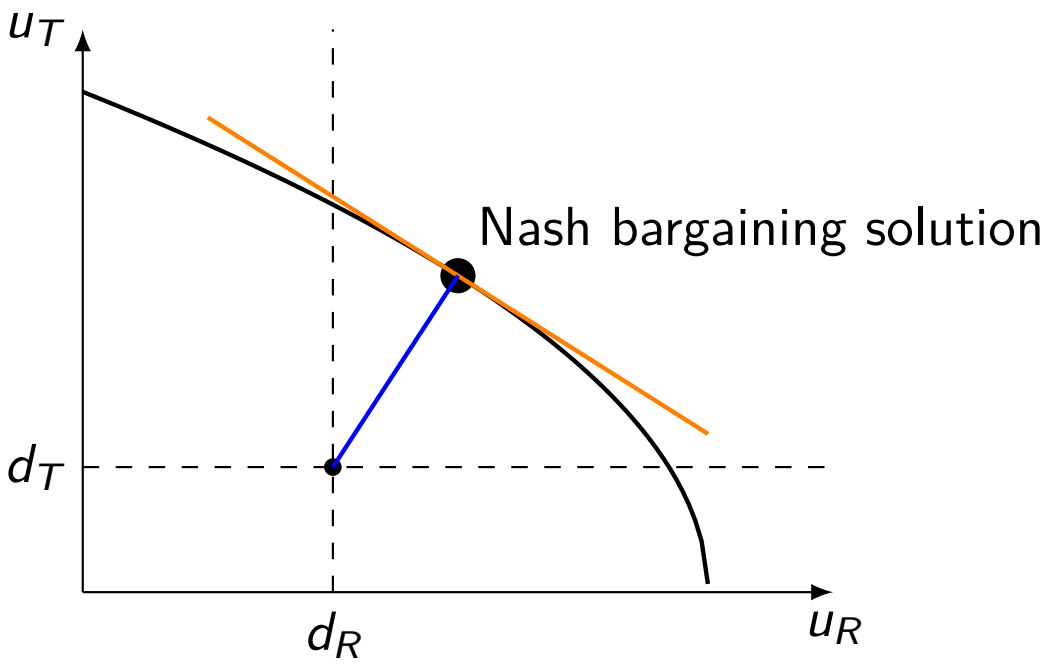
- Frontier is obtained by solving the below maximization: for every u_R

$$\max_{\theta \in \Theta} \mathbb{E}\left[U_T(J + S - R(S;\theta))\right]$$
$$\text{s.t. } \mathbb{E}\left[U_R(I + R(S;\theta))\right] \geq u_R.$$

- Issue: R includes a kink, making it challenging to solve analytically
- Solution:
  - Affine contract and mixed-type contract have the same bargaining set
  - Solve the same maximization in the space of affine contract
- Byproduct: the model is *incomplete*
  - There are the set of equilibrium mixed-type contracts
  - All we can get is the surplus split in the equilibrium
- The bargaining frontier

$$F(u_R) = -e^{-\rho_T\left(I+\mu-\frac{\rho_T\rho_R}{\rho_T+\rho_R}\frac{\sigma^2}{2}\right)}\left(-u_R\right)^{-\frac{\rho_T}{\rho_R}}.$$

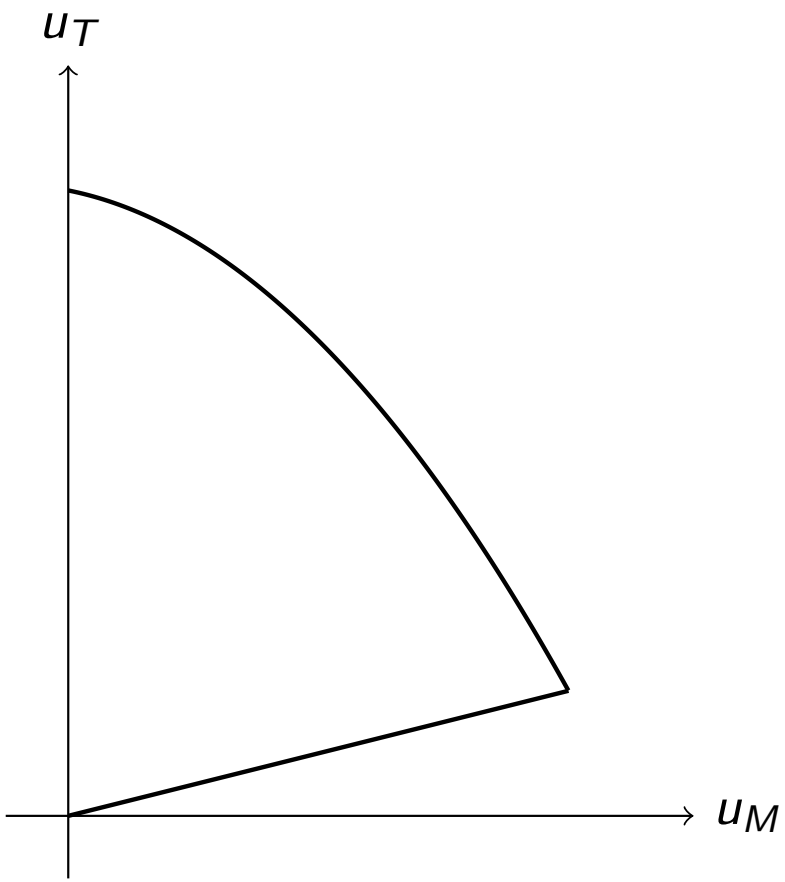
Risk Sharing: Nash Bargaining Solution



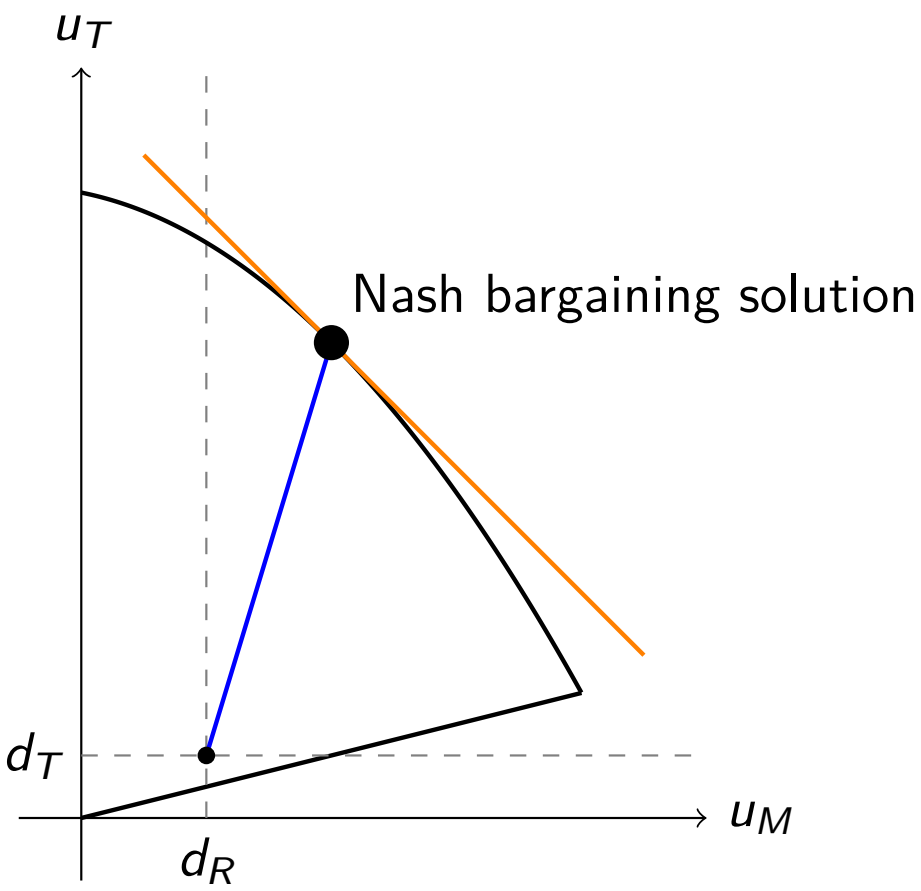
- Nash bargaining solution is determined by three objects:
  - Breakup point: $d_R = -e^{-\rho_R cI}$, $d_T = -e^{-\rho_T 0} = -1$
  - Frontier specified for each μ
  - B^{RS}: Assumption $B^{RS} = \frac{\rho_T}{\rho_R}$ (Roth and Rothblum, 1982)
- Both R and T like higher $\mu \rightarrow$ Over optimistic contract if fully delegated
  - The authority over μ is left to the manager, M

Effort Enforcement: Bargaining Frontier

- Expected utility of manager: $u_M(\mu) \equiv u_R(\mu) - \lambda \left(u_R(\mu) - \hat{u}_{NR}\right)^2$
- Bargaining set = Locus: $\{(u_M(\mu), u_T(\mu)) \mid \mu \in \mathbb{R}_+\}$
  - For any $\lambda > 0$, there is some $\underline{\mu}$ such that $\frac{d}{d\mu} u_M(\mu) < 0$ when $\mu \geq \underline{\mu}$



Effort Enforcement: Nash Bargaining Solution



- Breakup point is set to the same as before
- B^{EE} is exogeneously given (no assumption on the value)
- The model has explicit form of the surplus split: u_T, u_R and u_M

Continuation Decision

- Representative decides either termination or continuation
- Expected surplus from renewal vs Value of outside option

$$\mathbb{E}\left[u_R \mid \psi\right] > -e^{-\rho_R(cl+\psi)}.$$

  - ψ : monetary value of outside option
  - Expectation is taken w.r.t. B^{EE}: information gap between M and R

Value of Outside Option

- Search intensity (d): the count of the meeting with potential tenants
- d affects ψ

$$\psi = \tilde{\psi} + \beta_o d$$

  $\tilde{\psi}$ is the baseline value of outside option

- Endogeneity: Better outside option might decrease the search intensity
  - $\tilde{\psi}$ and d are simultaneously determined
  - No full model, just a correlation structure $\rightarrow$ *Control function approach*

Empirical Strategy

Tobit model

Variables

- An ongoing contract is denoted by a pair of i (tenant) and k (mall)
- Any contract is numerated with the contract number τ starting from 1
- Contract period (in months) is denoted by $T_{ik\tau}$
- Performance is measured for every months: $\tilde{X}_{ikt}$ for $t \in \{1, \cdots, T_{ik\tau}\}$
  - Main measure: *rank of sales per area within a mall*
- For an ongoing contract, we make a measure of performance:

$$X_{ik\tau} = \frac{1}{T} \sum_{t=1}^{T_{ik\tau}} \tilde{X}_{ikt}$$

  - the average of the performance measure during the contract period

Parametrization: Bargaining Power Ratio

- Logarithm of bargaining power ratio

$$\ln B_{ik\tau}^{UR} = X'_{ik\tau}\gamma + \varepsilon_{ik\tau}^{EE},$$

  - three variables relating with tenant; such as *rank of sales per area*
    - *area of the tenant*
    - *average of the previous sales per area*
    - *average of the previous ranking of sales per area*
  - Four variables depending on the shopping mall; such as *mall total sales*
    - *average of the previous mall total sales*
    - *average of the total number of tenants in the previous lease*
    - *average of the total number of customers in the previous lease*
    - *average of the monthly new tenant searches in the previous lease*
  - $\varepsilon_{ik\tau}^{EE}$ is unobserved factor

Parametrization: Value of Outside Option

- ψ basically captures the market demand for the retail space
- ψ is determined by three parts:
  - public information regarding the retail space, Z
  - search behavior conducted by the shopping mall, d
  - disturbance capturing the market's unobserved demand, $\tilde{\epsilon}^o$
- W : IV for search intensity $\rightarrow$ tenant specific performance measure
- Linear model (Petrin and Train, 2010, Wooldridge, 2015):

$$\psi_{ik\tau} = \left(Z'_{ik\tau}\gamma_o^{\psi} + \varepsilon_{ik\tau}^o\right) + \beta_o d_{ik\tau}$$
$$d_{ik\tau} = Z'_{ik\tau}\gamma_o^d + W'_{ik\tau}\delta + \nu_{ik\tau}^o$$
$$\varepsilon_{ik\tau} = \kappa\nu_{ik\tau}^o + \tilde{\varepsilon}_{ik\tau}^o$$

  - $\tilde{\varepsilon}_{ik\tau}^o$ is an exogeneous shock to the value of outside option
- Two random terms are i.i.d. joint Normal: $(\epsilon^{EE}, \tilde{\varepsilon}^o) \sim N(0, \Sigma)$

Estimation

- Estimation is conducted separately for the two shopping malls
- Two steps:
  1. Control function approach to make "reduced form" of outside option
  2. Tobit model to estimate all the parameters

Estimation: Control Function Approach

- Regression: $d_{ik\tau} = Z'_{ik\tau}\gamma_o^d + W'_{ik\tau}\delta + \nu_{ik\tau}^o$
- Obtain residual $\hat{\nu}_{ik\tau}^o$
- "Reduced form" equation for ψ is

$$\psi_{ik\tau} = Z'_{ik\tau}\gamma_o^\psi + \beta_o d_{ik\tau} + \kappa\hat{\nu}_{ik\tau}^o + \tilde{\varepsilon}_{ik\tau}^o$$

  - Treat $\hat{\nu}_{ik\tau}^o$ as an observed variable

Estimation: Tobit Model

- Likelihood function is constructed as in Tobit model:

$$\text{Likelihood} = \begin{cases} Pr(\text{observe contract } \theta_{ik\tau}) & \text{if continuation} \\ Pr(\text{exit}) & \text{otherwise} \end{cases}$$

Go to likelihood function

- Maximize log-likelihood under the constraint of Nash bargaining solution
  - Expected sales $\mu_{ik\tau}$ is endogeneous

Identification

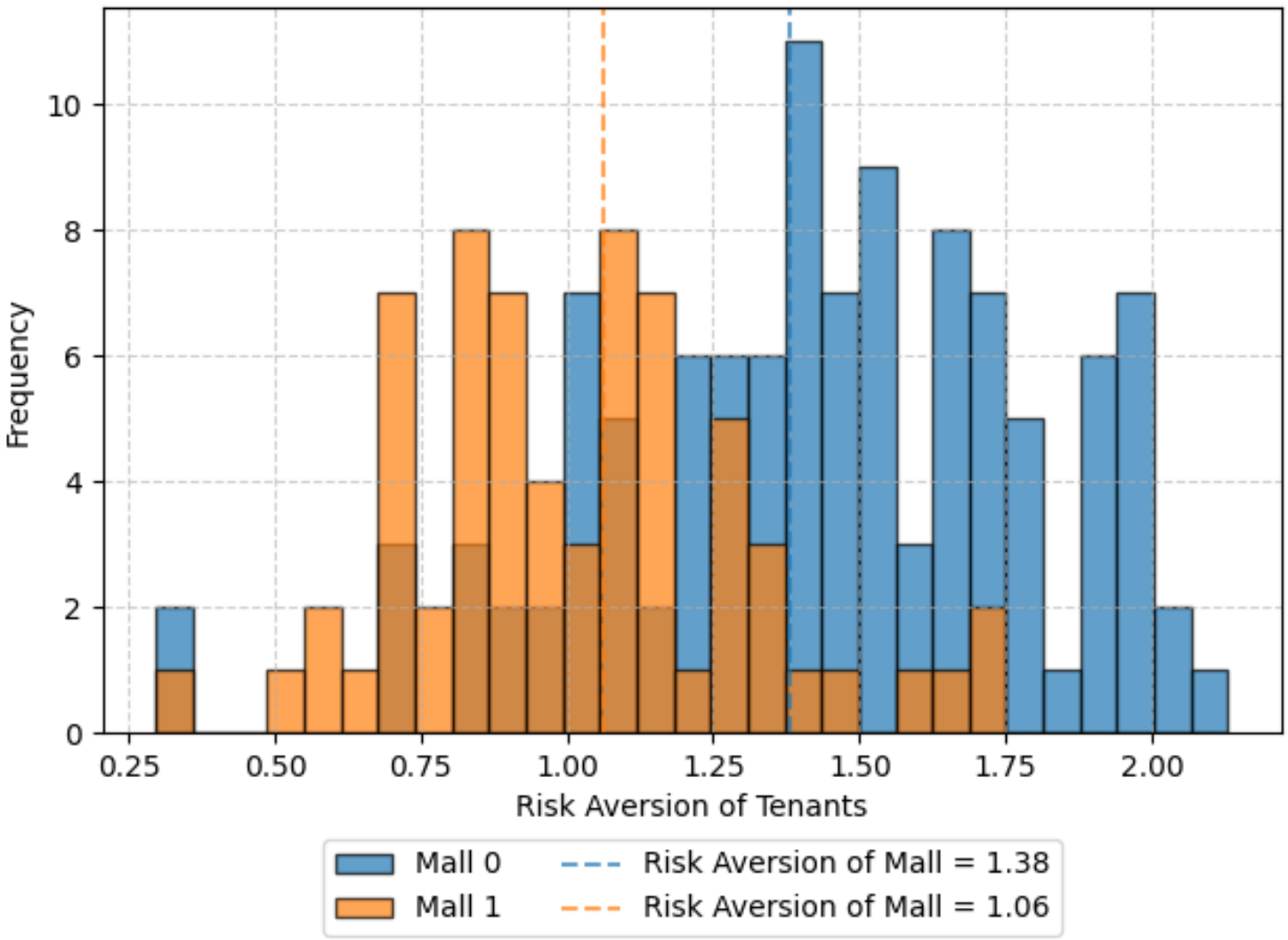
- Two sets of parameters
  1. Common across tenants within a mall
     - risk aversion of mall and marginal effects of covariates etc
  2. Contract-specific
     - Risk aversion of tenant: $\rho_{i\tau}$
     - Agreed upon expected sales (endogeneous parameter): $\mu_{ik\tau}$
- The first set is identified by the variation of exit offer
  - The second group is removed from the continuation decision
  - WHY?
    - Assumption $B^{RS} = \frac{\rho_T}{\rho_R}$ eliminates ρ_R from the decision
    - Cooperative approach allows me to parametrize $B^{EE}_{ik\tau}$ determining $\mu_{ik\tau}$
- The second set is identified by conditions of Nash solution
  - Two agents $\rightarrow$ Two surplus expressions $\rightarrow$ Two conditions

Results

Time-varying bargaining power and determinants of contract term

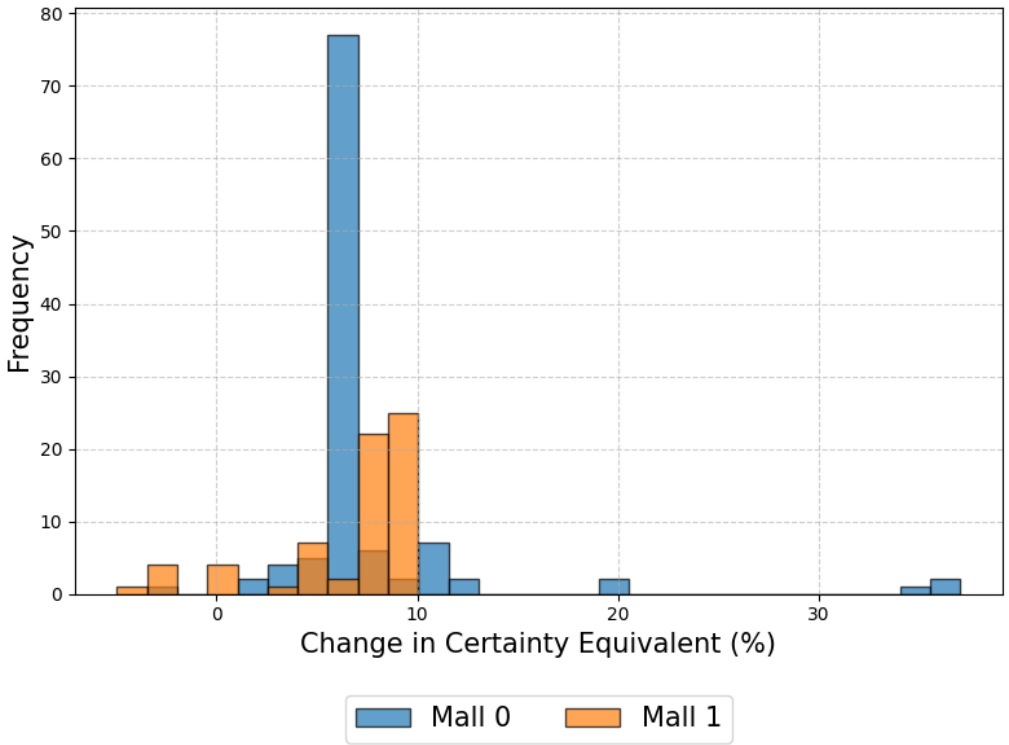
Risk Aversion

- Mall 0's manager is more risk averse



Influence from the Performance

- Rank of sales per unit area in a mall $\uparrow \rightarrow$ Tenant's bargaining strength $\uparrow$
  - Balance of bargaining powers varies over time
- Remove all dependencies on the past information from bargaining power
  - Remaining: Constant, Area, Risk aversions etc...
- Under CF bargaining positions, Mall's surplus increases by about 10%



Selection Mechanism of Contract Term

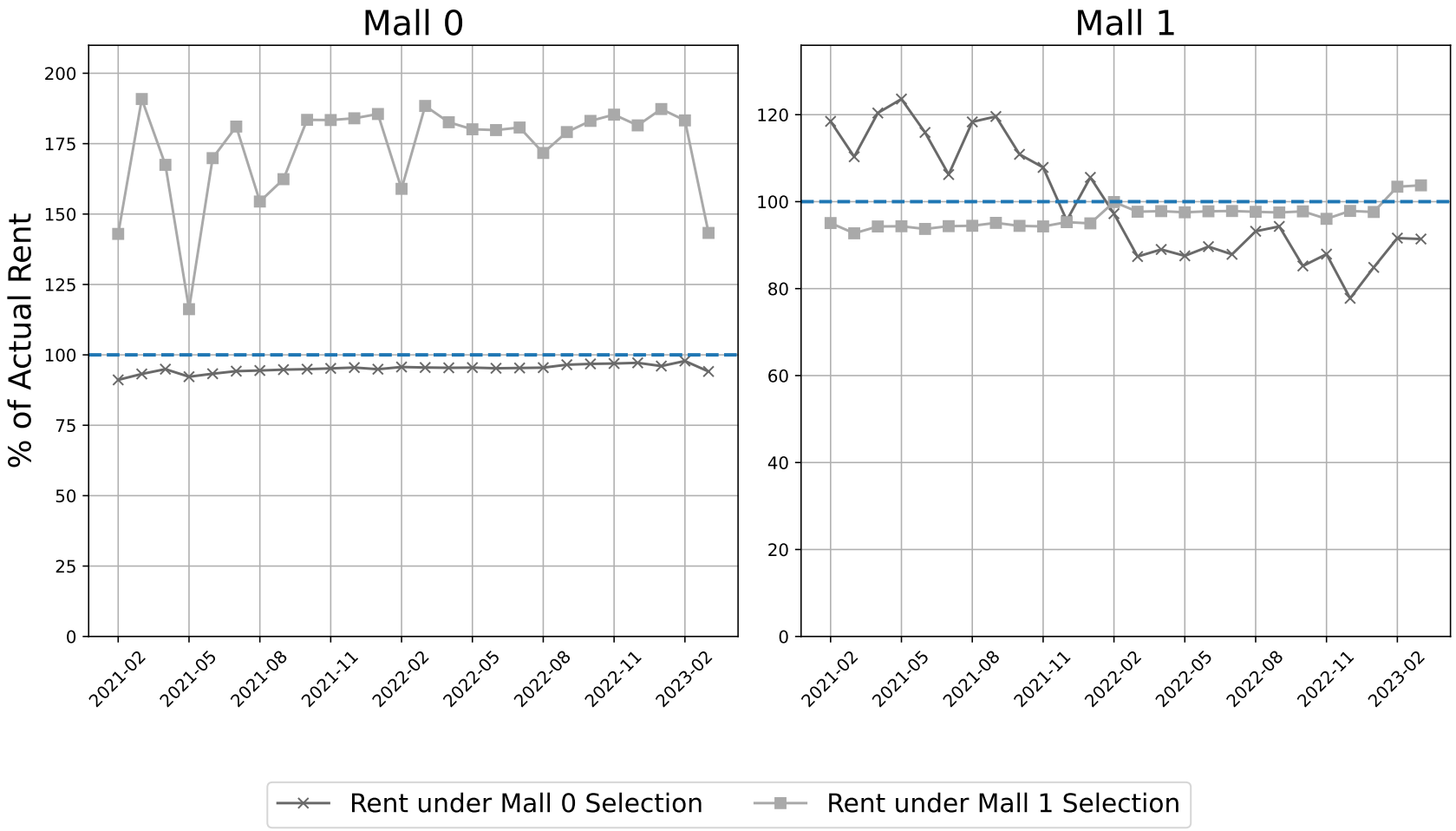
- Empirical analysis on contract term = Identifying the selection mechanism
- SUR model for the three parameters, Fixed, Base, and Rate

$$\begin{pmatrix} Fixed \\ Base \\ Rate \end{pmatrix} \sim \text{Bargaining Powers} + \text{Covariates} + \begin{pmatrix} \epsilon_f \\ \epsilon_b \\ \epsilon_r \end{pmatrix}$$

- Findings:
  - Bargaining power matters: Mall's power $\uparrow \rightarrow$ Base $\downarrow$ and Rate $\downarrow$
  - Mall 0 and Mall 1 differ in selection mechanism
    - Mall 0: Large volatility $\rightarrow$ Higher Fixed
    - Mall 1: Higher average sales $\rightarrow$ Higher base and Higher rate
    - Mall 0 gradually increases Rate
    - Mall 1 gradually increases Fixed

Counterfactual Rent

- Switch the selection mechanism between Mall 0 and 1
- Mall 0 can double the rent just by switching the manager



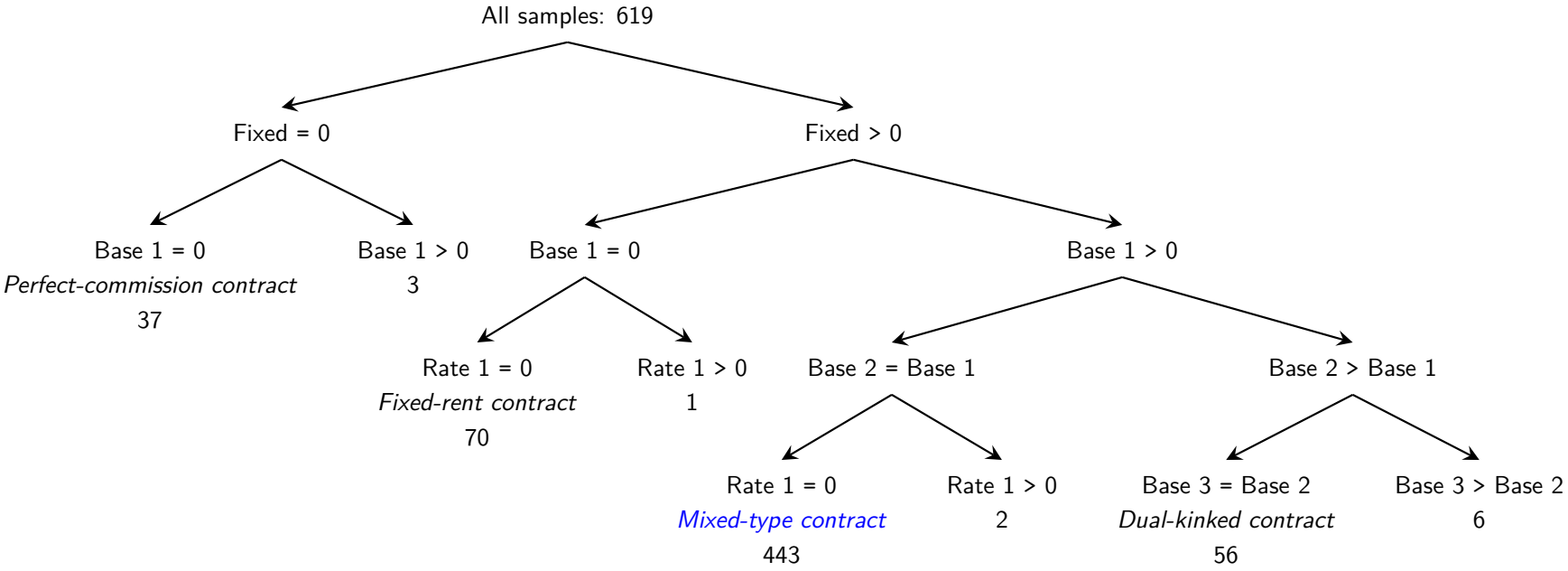
Conclusion

- Research question: How can we understand contract evolution?
- Propose a model of bargaining over contract
- Empirical challenge: Unobserved component of agreement
- Solution: Two-stage sequential bargaining
  - which is endogeneously supported
- Estimate the model using new data on tenant leasing in shopping malls
- Show the existence of influence from the past
- Identify the determinants of contract terms: Risk attitude matters

Appendix

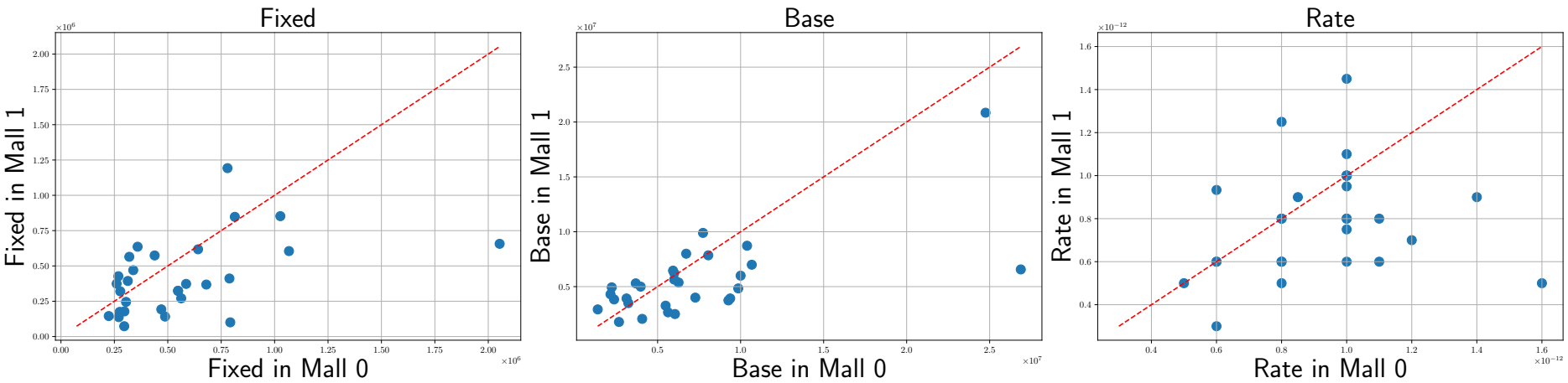
All Rent Structures

back to mixed type contract



Common Brands

back to descriptive stats



Contract Type	1	2	3	5	6	7	8
Mall 0	2	0	10	58	2	1	2
Mall 1	7	1	11	44	0	7	0

Table: Distribution of Rent Structures of Common Brands by Malls

Exit Offer

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Initial Offer \ Next Contract	**exit**	**same**	**up**	**down**	**updown**	**TOTAL**
exit	52	3	1	2	2	60
same	15	137	11	13	13	189
up	6	60	49	5	6	126
down	5	6	5	11	5	32
updown	1	1	3	5	9	19
TOTAL	79	207	69	36	35	426

Table: Initial Offer and Resulting Contract

Note: The table presents the relationship between the first initial offer (rows) and the resulting contract type (columns). Both the initial offer and the resulting contracts are categorized into five groups: exit, same, up, down, and updown. The numbers in the cells indicate the count of contracts corresponding to each combination of initial offer and resulting contract.

Likelihood Function

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- $\chi_{ik\tau} \in \{0, 1\}$ denotes continuation dummy
- Observation: $\mathcal{D}_{ik\tau} \equiv (\chi_{ik\tau}, \theta_{ik\tau}, X_{ik\tau}, Z_{ik\tau}, I_{ik\tau})$
- Parameters for mall k: $\xi_k \equiv \left(\gamma, \gamma_o^{\psi}, \beta_o, \Sigma, c, \lambda, \rho_k, \{\rho_{ik}\}_i\right)$

Let

$$u_M\left(\mathbf{X}, \epsilon^{EE}\right) = u_M(e^{\mathbf{X}'\gamma + \epsilon^{EE}}).$$

The likelihood for a tenant which exits is computed as follows:

$$L\left((\chi = 0, \phi), \mathbf{X}, \mathbf{Z}, I; \xi\right) = 1 - Pr\left(\mathbb{E}\left[u_M(\mathbf{X}, \epsilon^{EE}) \mid \tilde{\epsilon}^o\right] > -e^{-\rho_k\left(cI + \mathbf{Z}'\gamma_o^{\psi} + \beta_o d + \kappa\hat{\nu}^o + \tilde{\epsilon}^o\right)}\right)$$

where the probability is taken with $\tilde{\epsilon}^o$ and the expectation is taken with respect to ϵ^{EE}.

Likelihood Function

The likelihood for a continuing tenant is

$$
\begin{aligned}
&L\left((\chi = 1, (f, b, r)), \mathbf{X}, \mathbf{Z}, l; \xi\right) \\
&= \int_{\mathbb{E}[u_M(\mathbf{X}, \epsilon^u) | \tilde{\epsilon}^o] > -e^{-\rho_k\left(cl + \mathbf{Z}'\gamma_o + \beta_o d + \kappa \hat{\nu}^o + \tilde{\epsilon}^o\right)}} f\left(\hat{\epsilon}^{EE} \mid \tilde{\epsilon}^o\right) \times f(\tilde{\epsilon}^o) \mathrm{d}\tilde{\epsilon}^o,
\end{aligned}
$$

where the recovered value of unobserved term is computed as follows:

$$
\hat{\epsilon}^{EE} \equiv \ln\left(-\frac{1 - \lambda\left(u_M + c_M - 2\tilde{u}_M\right)}{1 - 2\lambda\left(u_M - \tilde{u}_M\right)} \frac{u_T}{c_T} \frac{c_M}{u_M}\right) - \mathbf{X}'\gamma.
$$

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